Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral of standard functions - Type 3

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Particular integral of standard functions - Type 3

1 Particular integral of standard functions - Type 3

Study focus: Particular integral of standard functions - Type 3. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We are solving a non-homogeneous differential equation of the type f of d of y is equal to capital X, where capital X is equal to x to the power n and is a positive integer.

Key details

- In the previous sessions, we learned methods or rules to find the particular integral of exponential function e to the power ax or trigonometric functions of the type sine ax plus b.
- R cos c plus b.
- So the step the first step being in f of d.
- Take out the lowest d degree common to get the expression in the form 1 plus some function of d which I call it as phi of d or 1 minus phi of d.
- And you are when you are expanding you are writing the terms up to d to the par n because the d to the power n plus 1 of x to the power n, that is n plus 1 third derivative of x to the po n is anyway 0, n plus 2 third.
- Derivative of x to the power n is 0.
- Or 1 minus 5 of d to the power minus 1, you write the terms up till d to the power n, neglecting the higher powers of d.
- Then operate on the numerator term by term by taking capital D is equal to the n to d by dx, d square as d2 by dx to d three d cube as d cube by dx cube and so on.
- The following expressions will be helpful to expand 1 plus phi of d to the power minus 1 or 1 minus phi of d to the power minus 1 in ascending powers of d.
- What is the first equation?
- 1 plus x whole to the power minus 1 is 1 minus.
- Minus x, plus x square, minus x cube, and so on.
- Here in the series, plus and minus sign alternates.

- We need not have to remember that.
- 1 minus x whole to the power minus 1 is 1.
- Minus x, which is plus x.
- Plus minus x the whole square, which is plus x square.
- Minus into minus x cubed, which is plus x cube, and so on.
- Similarly, the next expression may also be used.
- to expand this 1 plus 5 d to the power minus 1 or 1 minus 5 d to the power minus 1.
- 1 plus x whole to the power minus 2, which is 1 minus 2x plus 3x square minus 4x cube.
- See look here in this series also plus and minus sign alternates in the C.
- Then 1 minus x whole to the power minus 2.
- We need not have to remember this.
- So if we remember the first and the third, it is sufficient.
- So the second and the fourth can be derived from the first and the third.
- Find the general solution of d2y by dx2 minus y is equal to 5x minus 2.
- Look at the right hand side function is an algebraic function, 5x minus 2.
- So now because the equation is a non-homogeneous linear differential equation, so the general solution has two components: complementary function and particular integral.
- So the first step being the to find the complementary function.
- For that, first let me first write this in operated notation form capital D square minus 1 of y is equal to 5x minus 2.

Example or application from the lecture

- First particular integral is given by 1 over f of d of x to the power n.
- Let us consider a question.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

M square minus 1 is equal to 0 because the degree of this equation is 2.

Key details

- 2 roots, the two roots are plus or minus 1.
- What is the first step?
- Take out the lowest d degree term to get the expression in the form 1 plus phi d or 1 minus 5 d.
- Because minus 1 can be written as minus 1 into d to the power 0.
- Minus of 1 minus d square to the power 1 minus d square.
- That is the first step.
- I wrote it in the form 1 minus 5 of d to the power minus 1 of the function 5x minus 2.
- Now what is the next step?
- Using binomial theorem, expand this in ascending powers of D.
- 1 minus x whole power minus 1, which is 1 plus x plus x square plus x cube.
- Use that formula and expand it in ascending powers of d.

- I get it as minus of 1 minus d square plus d to the power 4.
- See this is 1 minus x whole to the power minus 1, which is 1 plus x plus x square plus 1.
- Before it was 1 minus x, so that is wrong.
- 1 plus x square plus x to the power 4 plus on because it is 1 minus x whole raised to minus 1.
- When I simplify this, see 1 into 5x minus 2 is 5x minus 2, and d square of 5x minus 2.
- What is d square of 5x minus 2?
- 2 times differentiation of 5x minus 2 with respect to x.
- First time differentiation of this 5x minus 2 itself is a constant.
- First order derivative of 5x minus 2 is 5.
- If I differentiate the constant again, I get 0.
- So this simplifies to minus into 5x minus 2 because these when I operate these d terms.
- terms on 5x minus 2 I am getting 0.
- What is particular integral?
- It is 2 minus 5x.
- Therefore, the general solution is y is equal to the complementary function which is denoted as yc plus the particular integral y p.
- Find the particular integral of d square plus phi d plus 4 of y is equal to x square plus 7x plus 9.
- See in this question I need not have to find the complementary function because the question is only to find the particular integrals.
- Considered here an algebraic function of degree 2.
- What is the first step?
- In f of d you take out the lowest d degree term as common factor and get this expression in the form 1 plus 5d or 1 minus 5d.
- You if you take 4 outside, it becomes 4 into 1 plus d square plus 5 d by 4.
- See this is what we get.
- Of x square plus 7x plus 9.

Example or application from the lecture

- Particular integral is given by x square plus 7x plus 9 divided by d square plus by d plus 4.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

If you look at the expression over here, algebraic expression, it is of degree 2.

Key details

- minus this term that is d square plus phi d by four plus.
- X square that is d square plus y d by 4 the whole square.
- I am neglecting the higher power sub d because d3, d4, dn of x square plus 7x plus 9 is anyway equal to 0.
- The whole square.

- First use a plus b the whole square and you simplify.
- 1 minus d square by 4 minus 5 b by 4 plus in this a plus b the whole square.
- A square plus 2ab plus b square.
- When I differentiate, I get d square of x square plus 7x plus 9 as constant.
- Higher powers of d is anyway, when it is operated on x square plus 7x plus 9, I get 0.
- I have written only d square by 4 the.
- Sorry, 5 d by 4 the whole square, which is 25 d square by 60.
- I've neglected d square the whole square, that is d to the power 4 into 2 into 5 d into d, that is 10 d cube that I have neglected and I have considered only 5 d by 4 the whole square which is 25 by 5 25 d square by 16.
- See d of x square plus 7x plus 9 that is 1 time differentiation 2x plus 7.
- 2 times differentiation is 2.
- Higher derivatives will be anyway equal to 0.
- 1 by 4 into 1 into x square plus 7x plus 9 minus 1 by 4 into what is d square of this?
- It is equal to 2.
- Minus 5 by 4 into d of x square plus 7x plus 9 is 2x plus 7.
- Plus 20 by 16 into d square is 2.
- This is how we find the particular integraph of integral of algebraic function of the type x to the power n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral of standard functions - Type 3 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral of standard functions - Type 3 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Particular integral of standard functions - Type 3 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0050. Sidecar variants compare...

C Revision and retrieval practice